Medicine, Dentistry and Medical laboratory

Cell Biology

Lecture 6&7

Passage of Molecules into and out of the Cell

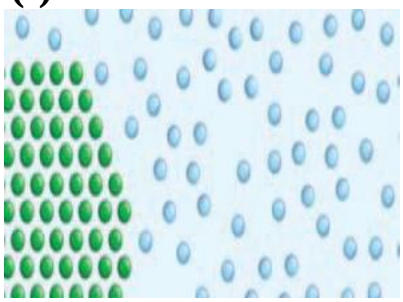
(A) Passive Transport Across Cell Membrane:

* Molecules can passively (no energy required) cross a membrane. In general, small, non-charged molecules, such as carbon dioxide, oxygen, glycerol, and alcohol, can freely cross the membrane.
* They are able to slip between the hydrophilic heads of the phospholipids and pass through the hydrophobic tails of the membrane.
* These molecules are said to follow their **concentration gradient** as they move from an area where their concentration is high to an area where their concentration is low.

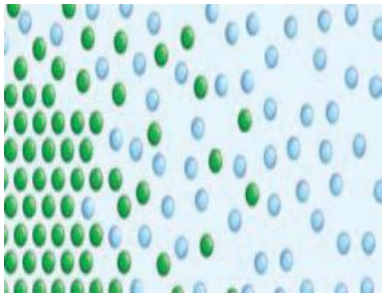
1- Diffusion:

* Is the movement of molecules from a higher to a lower concentration (down their concentration gradient) until equilibrium is achieved and they are distributed equally.
* Diffusion is a physical process due to random molecular motion that can be observed with any type of molecule.
* Several factors influence the rate of diffusion; temperature, pressure, electrical currents and molecular size.

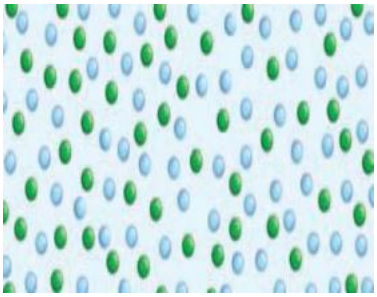
(I) Process of diffusion:



a. Crystal of dye is placed in water



b. Diffusion of water and dye molecules



c. Equal distribution of molecules results

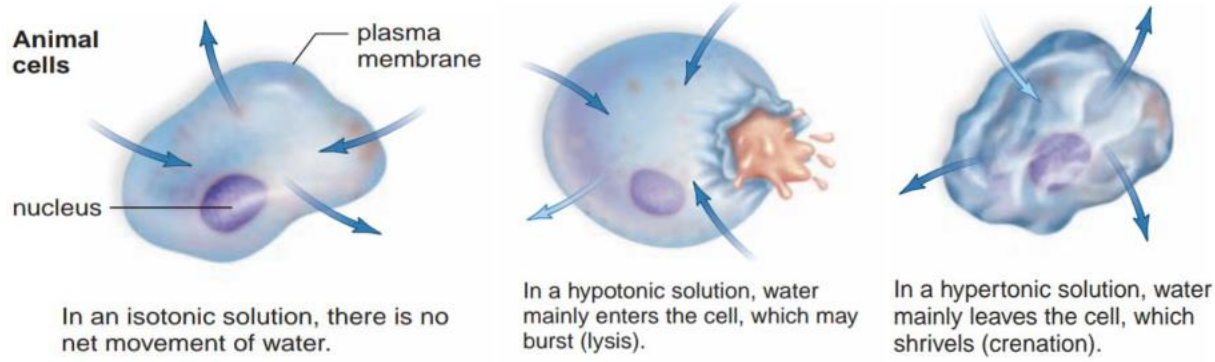
(II) Gas exchange in lungs:

Oxygen (O_2) diffuses into the capillaries of the lungs because there is a higher concentration of oxygen in the alveoli (air sacs) than in the capillaries.

2- Osmosis:

* It is the movement of **water molecules** from a region of higher water concentration to a region of lower water concentration through a partially permeable membrane.

* The diffusion of water across a differentially (selectively) permeable membrane due to concentration differences is called **osmosis.**



(I) Isotonic Solution:

In the laboratory, cells are normally placed in **isotonic solutions** (the solute concentration and the water concentration both inside and outside the cell are equal) and there is no net gain or loss of water.

* The prefix **iso** means "the same as," and the term **tonicity** refers to the strength of the solution.

(II) Hypotonic Solution:

Solutions that cause cells to swell, or even to burst, due to an intake of water are said to be **hypotonic solutions.**

***** The prefix **hypo** means "less than" and refers to a solution with a lower concentration of solute (higher concentration of water) than inside the cell.

* If a cell is placed in a hypotonic solution, water enters the cell; the net movement of water is from the outside to the inside of the cell.

(III) Hypertonic Solution:

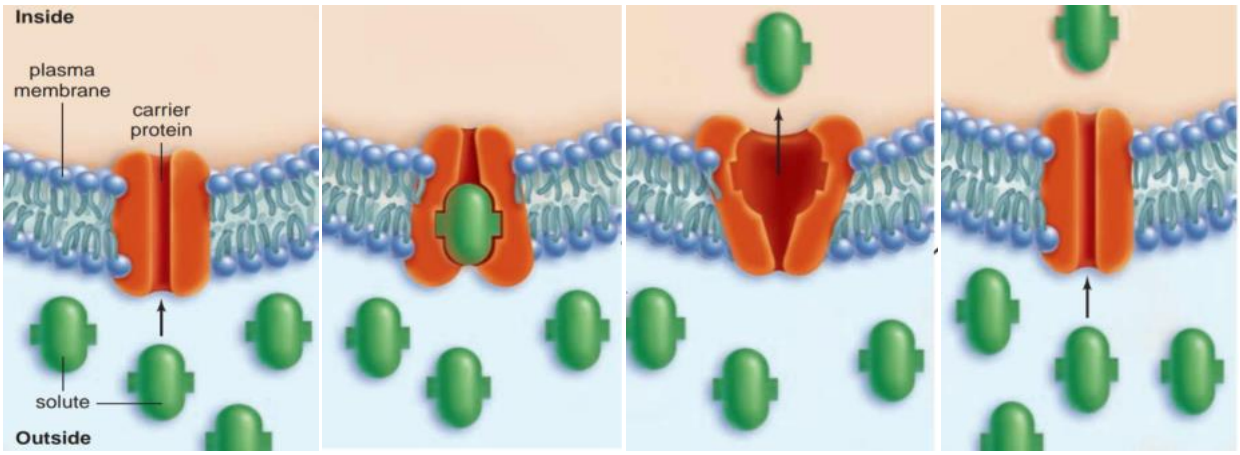
Solutions that cause cells to shrink or shrivel due to loss of water are said to be **hypertonic solutions.**

***** The prefix **hyper** means "more than" and refers to a solution with a higher percentage of solute (lower concentration of water) than the cell.

* If a cell is placed in a hypertonic solution, water leaves the cell; the net movement of water is from the inside to the outside of the cell.

3- Facilitated transport:

* Explains the rapid passage of water and also such molecules as glucose and amino acids across the plasma membrane. Whereas water moves through a channel protein, the passage of glucose and amino acids is facilitated by their reversible combination with carrier proteins, which transport them through the plasma membrane.



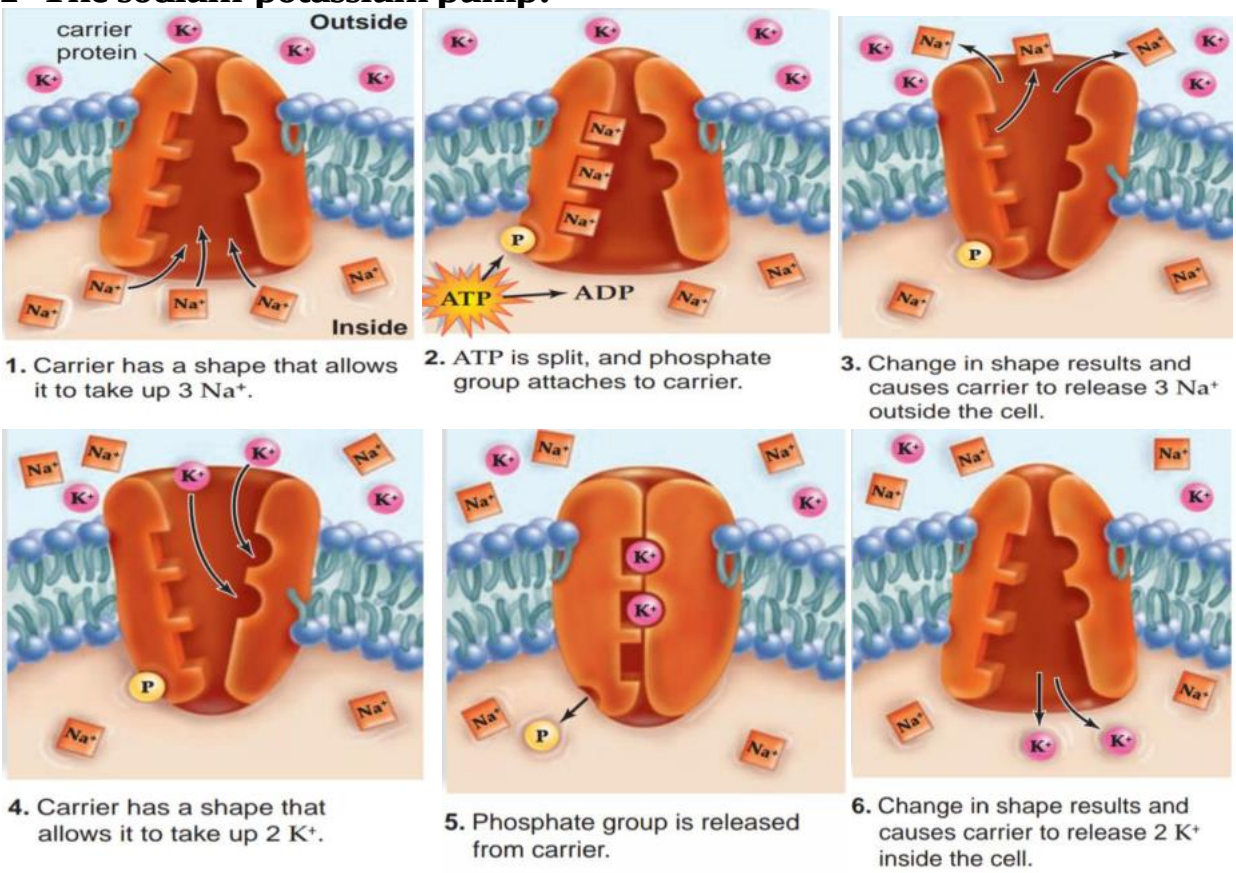
* These carrier proteins are specific. For example, various sugar molecules of identical size might be present inside or outside the cell, but glucose can cross the membrane hundreds of times faster than the other sugars. As stated earlier, this is the reason the membrane can be called differentially permeable.

* The carrier protein undergoes a change in shape as it moves a solute across the membrane.

(B) Active Transport Across A membrane:

* During **active transport,** molecules or ions move through the plasma membrane, accumulating either inside or outside the cell.
* For example, iodine collects in the cells of the thyroid gland; glucose is completely absorbed from the gut by the cells lining the digestive tract; and sodium can be almost completely withdrawn from urine by cells lining the kidney tubules.
* In these instances, molecules have moved to the region of higher concentration, exactly opposite to the process of diffusion.

1- The sodium-potassium pump:



1. Carrier has a shape that allows it to take up 3 Na^+.
2. ATP is split, and phosphate group attaches to carrier.
3. Change in shape results and causes carrier to release 3 Na^+ outside the cell.
4. Carrier has a shape that allows it to take up 2 K^+.
5. Phosphate group is released from carrier.
6. Change in shape results and causes carrier to release 2 K^+ inside the cell.

* The same carrier protein transports sodium ions (Na+) to the outside of the cell and potassium ions (K+) to the inside of the cell because it undergoes an ATP-dependent change in shape.
* Three sodium ions are carried outward for every two potassium ions carried inward; therefore, the inside of the cell is negatively charged compared to the outside.

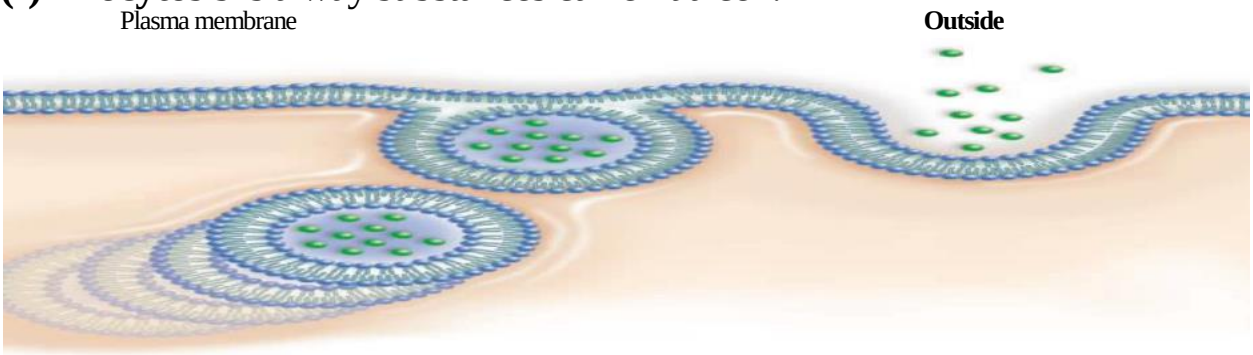
(2) Bulk Transport:

How do macromolecules such as polypeptides, polysaccharides, or poly nucleotides enter and exit a cell?

Because they are too large to be transported by carrier proteins, macromolecules are transported into and out of the cell by vesicle formation.

* Vesicle formation is membrane-assisted transport because membrane is needed to form the vesicle.

(I) Exocytosis is a way substances can exit a cell.



* Exocytosis, a vesicle fuses with the plasma membrane as secretion occurs. Cells of particular organs are specialized to produce and export molecules. For example, pancreatic cells produce digestive enzymes or insulin.

(II) Endocytosis is a way substances can enter a cell.

* During endocytosis, cells take in substances by vesicle formation.

* A portion of the plasma membrane invaginates to envelop the substance, and then the membrane pinches off to form an intracellular vesicle.

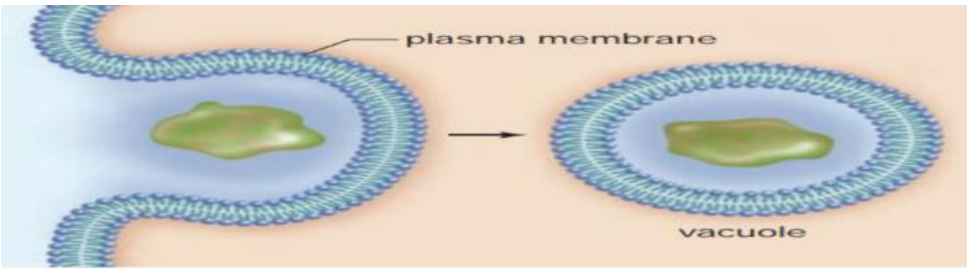
Endocytosis occurs in one of three ways:

(a) Phagocytosis.

(b) Pinocytosis.

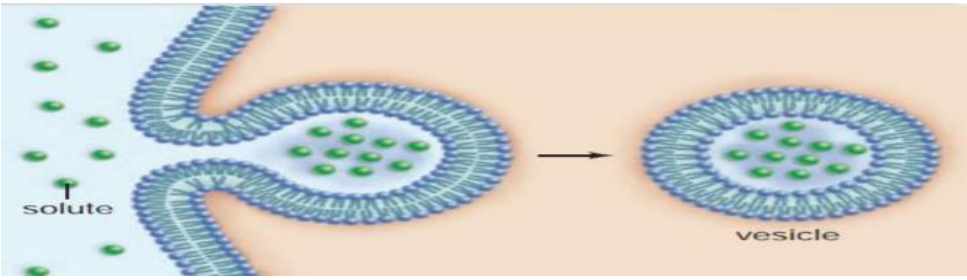
(c) Receptor-mediated endocytosis.

(a) Phagocytosis: transports large substances such as a food particle.



* Phagocytosis is common in unicellular organisms such as **amoebas** and certain types of human **white blood cells** are amoeboid.

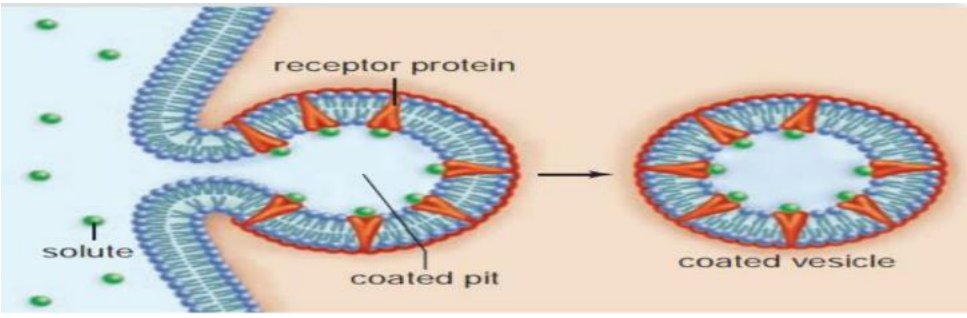
(b) Pinocytosis: transports small substances into a cell.



* Occurs when vesicles form around a liquid or around very small particles. Blood cells, cells that line the kidney tubules or the intestinal wall, and plant root cells all use pinocytosis to ingest substances

(c) Receptor-mediated endocytosis: is a special form of pinocytosis.

* It uses a receptor protein shaped in such a way that a specific molecule such as a vitamin and peptide hormone.



* Once a specific solute binds to receptors, a coated pit becomes a coated vesicle. After losing the coat, the vesicle can join with the lysosome, or after discharging the substance, the receptor-containing vesicle can fuse with the plasma membrane.